Order of Operations

Name _____ **Date** _____ No calculator Version 1

KEY WORDS order of operations · GEMA · brackets · exponent · evaluate

A Two operations ●○○ ANSWER ON THE SHEET

Remember: GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

1. Work out:

a) $3 + 4 \times 5 =$ _____ **b)** $20 - 12 \div 4 =$ _____ **c)** $6 \times 3 - 8 =$ _____ **d)** $18 \div 2 + 7 =$ _____

e) $5 + 30 \div 6 =$ _____ **f)** $24 - 3 \times 5 =$ _____ **g)** $7 \times 2 + 6 \times 3 =$ _____ **h)** $40 \div 8 - 12 \div 6 =$ _____

B Brackets ●○○ ANSWER ON THE SHEET

2. Work out:

a) $(3 + 4) \times 5 =$ _____ **b)** $20 \div (2 + 3) =$ _____ **c)** $2 \times (9 - 4) + 1 =$ _____

d) $(8 + 4) \div (5 - 2) =$ _____ **e)** $30 - (4 + 2 \times 3) =$ _____ **f)** $3 \times (7 - (2 + 1)) =$ _____

g) $(15 - 3) \div 4 + 6 =$ _____ **h)** $50 - 2 \times (11 + 4) =$ _____

C Exponents and roots ●●○ ANSWER ON THE SHEET

3. Work out:

a) $2 + 3^2 =$ _____ **b)** $5 \times 2^3 =$ _____ **c)** $(2 + 3)^2 =$ _____ **d)** $4^2 - 3 \times 5 =$ _____

e) $\sqrt{16} + 2 \times 5 =$ _____ **f)** $3 \times \sqrt{49} - 2^2 =$ _____ **g)** $(6 - 2)^2 \div 8 =$ _____ **h)** $10^2 - 6^2 =$ _____

D Negative numbers ●●○ ANSWER ON THE SHEET

⚠ **Watch out:** $(-3)^2 = 9$ but $-3^2 = -9$, because the exponent is done before the minus sign.

4. Work out:

a) $-4 + 3 \times 2 =$ _____ **b)** $6 - 2 \times 5 =$ _____ **c)** $(-3)^2 + 4 =$ _____ **d)** $-3^2 + 4 =$ _____

e) $-2 \times (5 - 9) =$ _____ **f)** $12 \div (-4) + 7 =$ _____ **g)** $(-6 + 2) \times 3 =$ _____ **h)** $5 - (-2)^3 =$ _____

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E Fractions as division ●●● WORK IN YOUR BOOK

Remember: A fraction bar works like brackets: work out the top and the bottom first, then divide.

5. Work out:

a) $\dfrac{6+14}{5}$ **b)** $\dfrac{3\times 8}{2+4}$ **c)** $\dfrac{5^2-1}{2\times 3}$

d) $7+\dfrac{18}{2\times 3}$ **e)** $\dfrac{4+6\times 5}{10-8}$ **f)** $\dfrac{(3+5)^2}{4^2}$

F Mixed practice ●●● WORK IN YOUR BOOK

6. Work out:

a) $4+2\times(3^2-5)$ **b)** $(8-12)\div 2+3^2$ **c)** $2^3\times 5-4^2\div 8$

d) $\dfrac{20-2\times 4}{3}+5$ **e)** $-2\times 3^2+20$ **f)** $(4+2)^2-(4^2+2^2)$

g) $100\div(2\times 5)^2$ **h)** $3+2\times(10-2\times 3)$ **i)** $\sqrt{3^2+4^2}\times 2$

7. Find the missing number.

a) $5+\square\times 3=17$ **b)** $(\square+2)\times 5=45$ **c)** $30-4\times\square=10$ **d)** $\square^2+6=55$

8. Write one calculation for each problem, then work it out.

a) Tickets cost \$12 for an adult and \$7 for a child. Find the cost for 3 adults and 4 children.

b) A taxi charges \$4 plus \$3 for each kilometre. Find the cost of a 9 km trip.

c) Sam buys 5 pens at \$2 each and pays with a \$20 note. Find his change.

G Brackets and mistakes ●●● WORK IN YOUR BOOK

9. Copy each calculation and put in one pair of brackets to make it true.

a) $3+4\times 5=35$ **b)** $20-8\div 4=3$

c) $2\times 3+5^2=56$ **d)** $36\div 4+2\times 3=18$

10. Tama says that $8-2\times 3=18$. What did he do wrong? What is the correct answer?

11. Aroha types $12+8\div 4$ into a calculator and gets 14. Hemi gets 5. Who is right, and what did the other person do?

★ Extension FOR EARLY FINISHERS

12. Use the numbers 2, 3, 4 and 5 once each, with any operations and brackets, to make 26.

13. Work out $\dfrac{(-2)^3+4\times 7}{\sqrt{25}}$.

14. Work out $\left(\dfrac{1}{2}\right)^2\times 12+2^3\div 4$.

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Order of Operations

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS order of operations · GEMA · brackets · exponent · evaluate

A Two operations ●○○ — ANSWER ON THE SHEET

Remember: GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

1. Work out:

a) $2 + 6 \times 4 =$ _____ **b)** $30 - 18 \div 3 =$ _____ **c)** $8 \times 2 - 9 =$ _____ **d)** $27 \div 3 + 5 =$ _____

e) $4 + 42 \div 7 =$ _____ **f)** $32 - 4 \times 6 =$ _____ **g)** $5 \times 3 + 4 \times 6 =$ _____ **h)** $36 \div 4 - 20 \div 5 =$ _____

B Brackets ●○○ — ANSWER ON THE SHEET

2. Work out:

a) $(2 + 6) \times 4 =$ _____ **b)** $24 \div (5 + 1) =$ _____ **c)** $3 \times (8 - 2) + 2 =$ _____

d) $(9 + 6) \div (7 - 4) =$ _____ **e)** $40 - (5 + 3 \times 4) =$ _____ **f)** $2 \times (10 - (3 + 2)) =$ _____

g) $(20 - 4) \div 8 + 7 =$ _____ **h)** $60 - 3 \times (9 + 6) =$ _____

C Exponents and roots ●●○ — ANSWER ON THE SHEET

3. Work out:

a) $4 + 2^3 =$ _____ **b)** $3 \times 4^2 =$ _____ **c)** $(1 + 4)^2 =$ _____ **d)** $5^2 - 4 \times 3 =$ _____

e) $\sqrt{25} + 3 \times 4 =$ _____ **f)** $2 \times \sqrt{81} - 3^2 =$ _____ **g)** $(7 - 3)^2 \div 2 =$ _____ **h)** $9^2 - 5^2 =$ _____

D Negative numbers ●●○ — ANSWER ON THE SHEET

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$, because the exponent is done before the minus sign.

4. Work out:

a) $-5 + 4 \times 3 =$ _____ **b)** $4 - 3 \times 6 =$ _____ **c)** $(-4)^2 + 3 =$ _____ **d)** $-4^2 + 3 =$ _____

e) $-3 \times (2 - 8) =$ _____ **f)** $20 \div (-5) + 9 =$ _____ **g)** $(-9 + 4) \times 2 =$ _____ **h)** $8 - (-3)^3 =$ _____

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E Fractions as division ●●● WORK IN YOUR BOOK

Remember: A fraction bar works like brackets: work out the top and the bottom first, then divide.

5. Work out:

a) $\dfrac{9+15}{6}$ **b)** $\dfrac{4\times 9}{2+4}$ **c)** $\dfrac{6^2-4}{2\times 4}$

d) $5+\dfrac{24}{3\times 4}$ **e)** $\dfrac{2+8\times 3}{15-13}$ **f)** $\dfrac{(2+4)^2}{3^2}$

F Mixed practice ●●● WORK IN YOUR BOOK

6. Work out:

a) $5+3\times(2^3-6)$ **b)** $(6-14)\div 4+2^3$ **c)** $3^2\times 4-6^2\div 9$

d) $\dfrac{30-3\times 4}{6}+7$ **e)** $-3\times 2^2+15$ **f)** $(5+1)^2-(5^2+1^2)$

g) $144\div(2\times 3)^2$ **h)** $4+3\times(12-2\times 5)$ **i)** $\sqrt{6^2+8^2}\times 3$

7. Find the missing number.

a) $4+\square\times 5=34$ **b)** $(\square+3)\times 4=44$ **c)** $40-6\times\square=22$ **d)** $\square^2+9=90$

8. Write one calculation for each problem, then work it out.

a) Tickets cost \$15 for an adult and \$8 for a child. Find the cost for 2 adults and 5 children.

b) A taxi charges \$5 plus \$4 for each kilometre. Find the cost of a 7 km trip.

c) Mia buys 6 pens at \$3 each and pays with a \$50 note. Find her change.

G Brackets and mistakes ●●● WORK IN YOUR BOOK

9. Copy each calculation and put in one pair of brackets to make it true.

a) $2+6\times 3=24$ **b)** $30-6\div 3=8$

c) $3\times 2+4^2=54$ **d)** $48\div 6+2\times 5=30$

10. Lani says that $10+6\div 2=8$. What did she do wrong? What is the correct answer?

11. Mere types $18-6\div 3$ into a calculator and gets 16. Jack gets 4. Who is right, and what did the other person do?

★ Extension FOR EARLY FINISHERS

12. Use the numbers 2, 3, 5 and 6 once each, with any operations and brackets, to make 28.

13. Work out $\dfrac{(-3)^2+5\times 3}{\sqrt{16}}$.

14. Work out $\left(\dfrac{1}{3}\right)^2\times 18+3^2\div 9$.

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